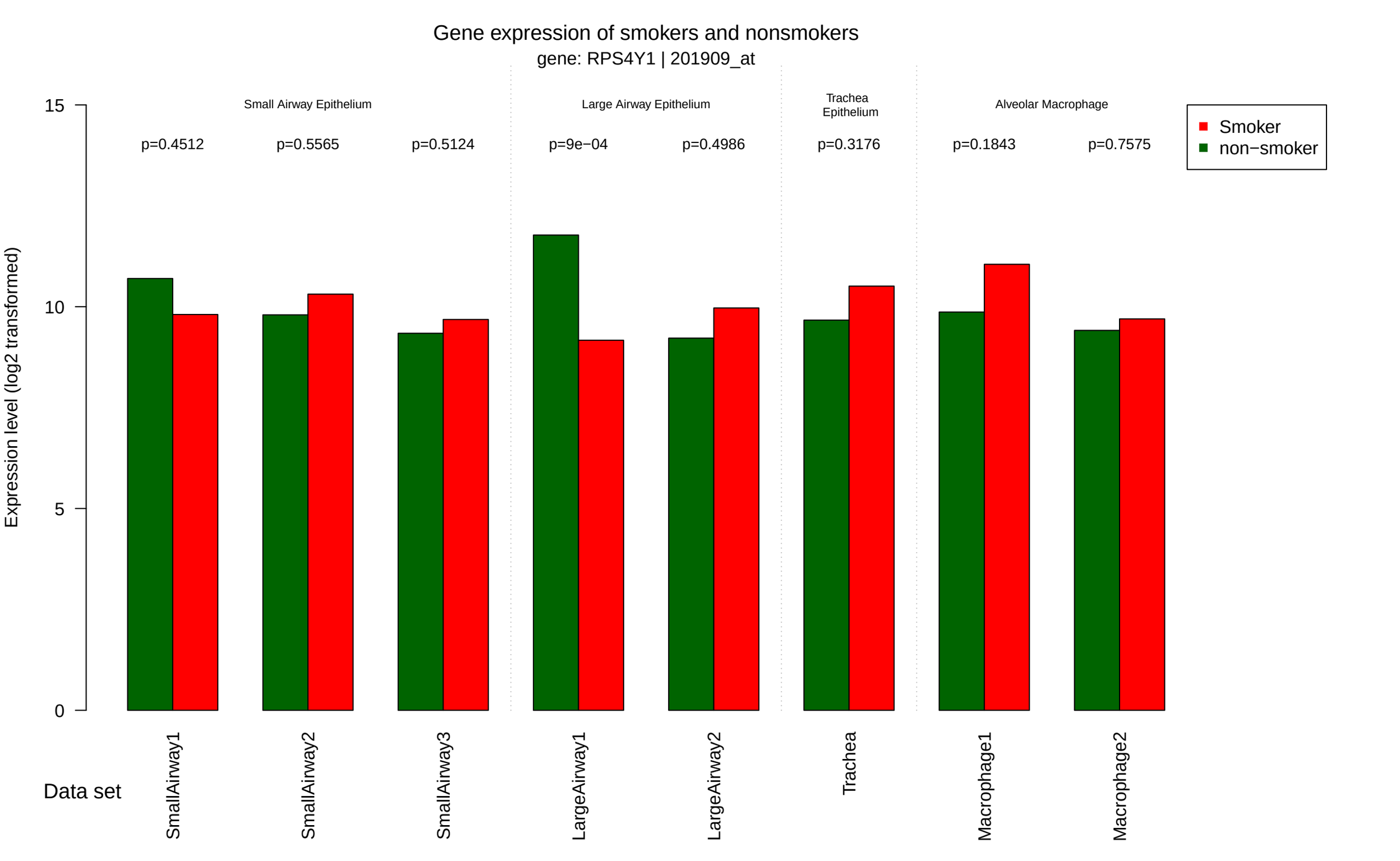
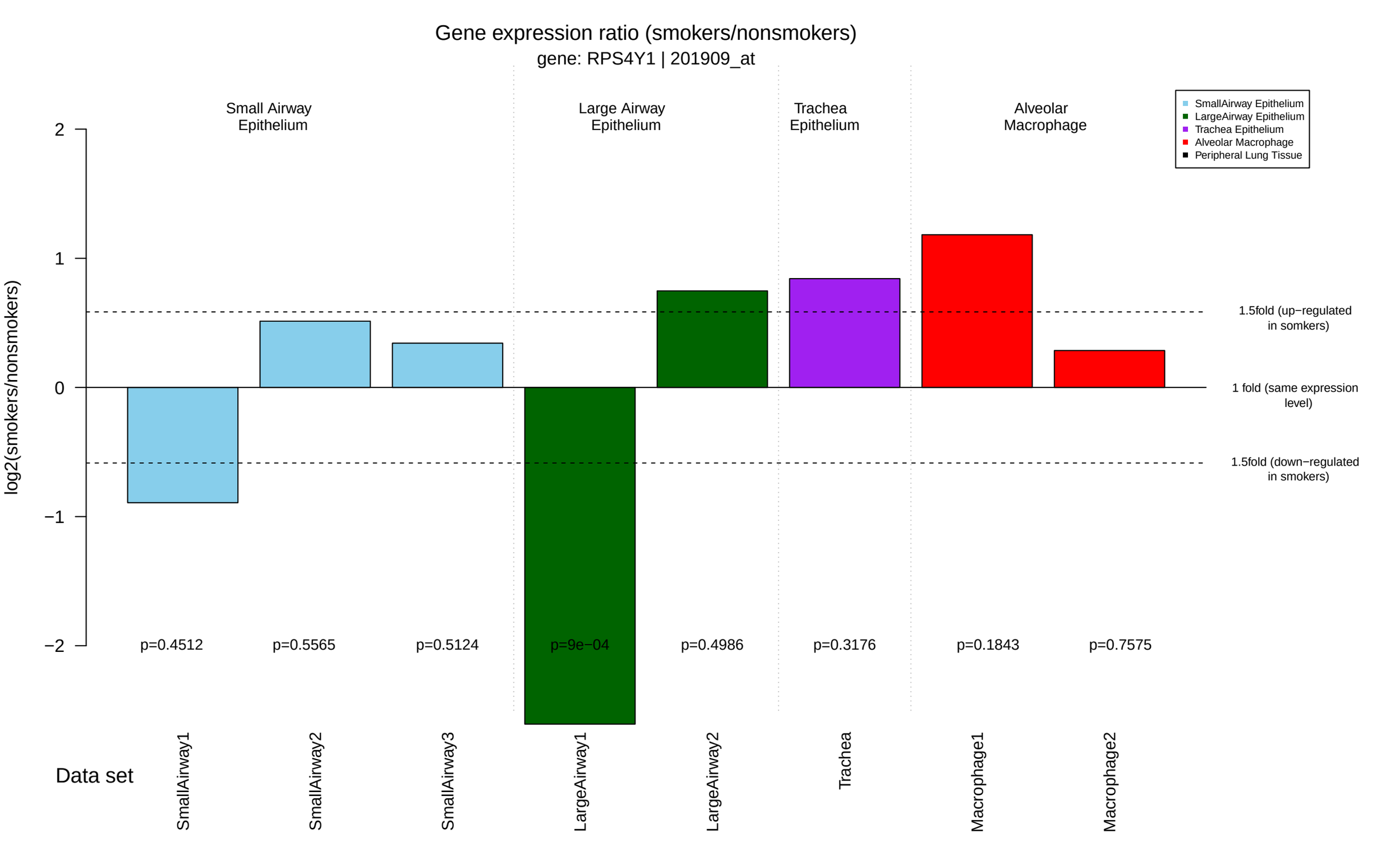
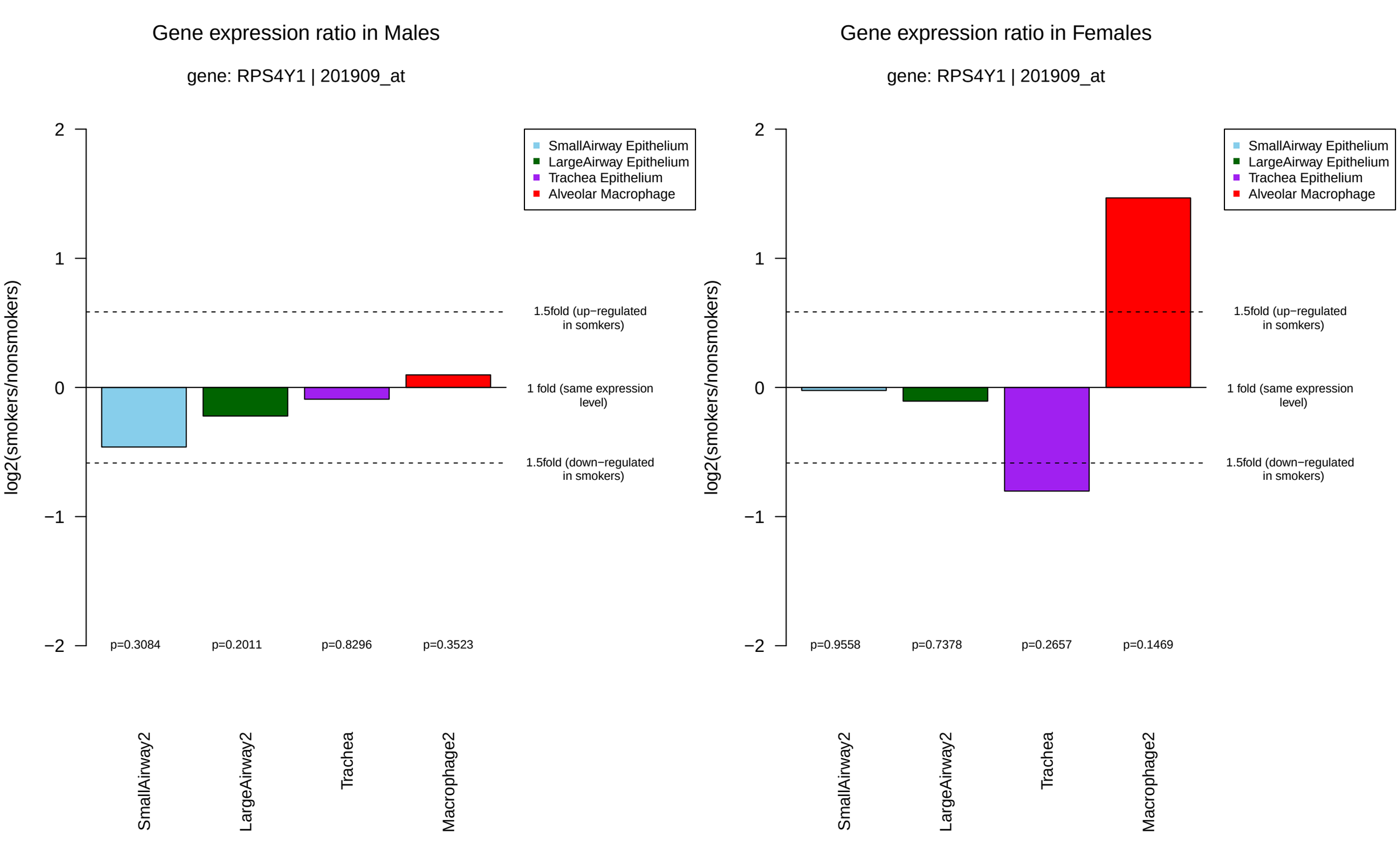
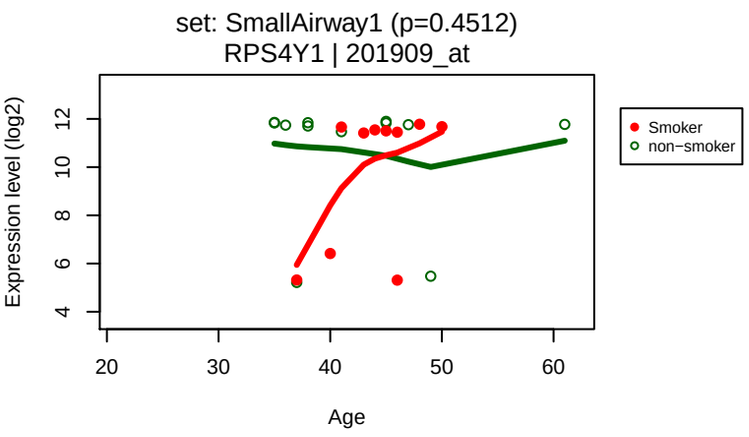
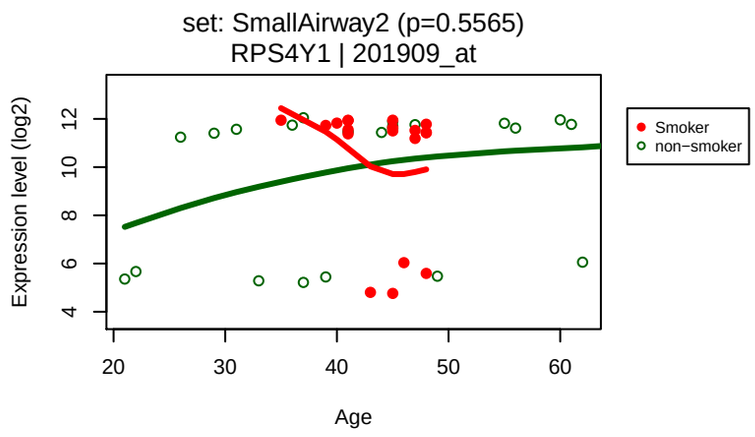
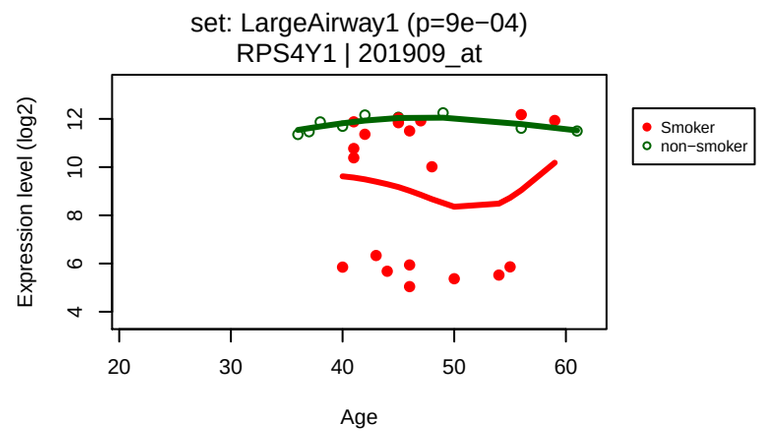

Gene expression of smokers and nonsmokers
gene: RPS4Y1 | 201909_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4512
p=0.5565
p=0.5124
p=9e−04
p=0.4986
p=0.3176
p=0.1843
p=0.7575
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

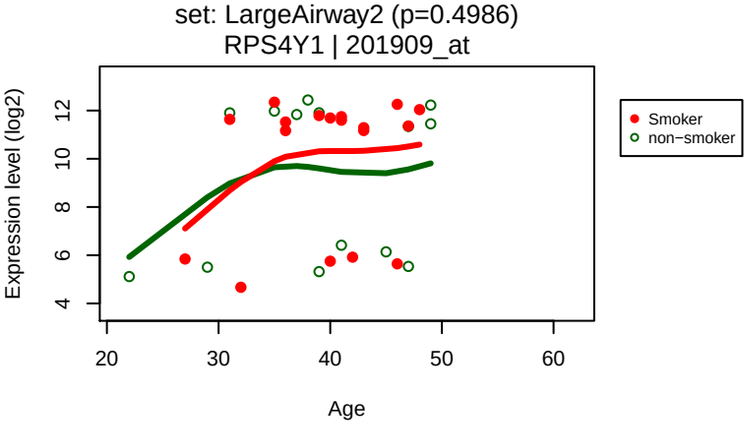

Gene expression ratio (smokers/nonsmokers)
gene: RPS4Y1 | 201909_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4512
p=0.5565
p=0.5124
p=9e−04
p=0.4986
p=0.3176
p=0.1843
p=0.7575
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

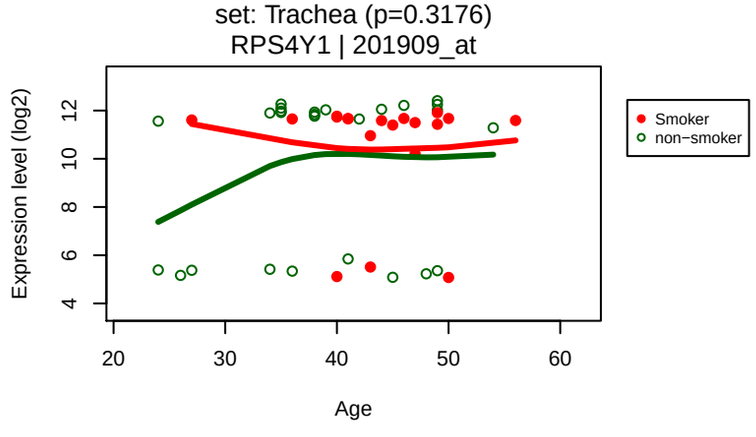

Gene expression ratio in Males
gene: RPS4Y1 | 201909_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3084
p=0.2011
p=0.8296
p=0.3523
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS4Y1 | 201909_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9558
p=0.7378
p=0.2657
p=0.1469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

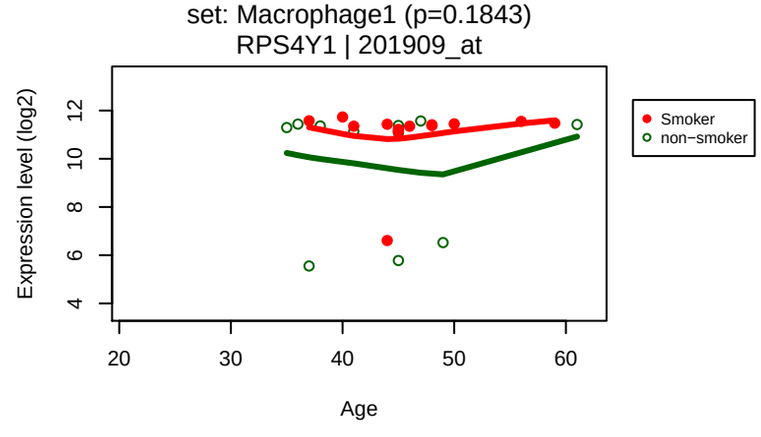

set: SmallAirway1 (p=0.4512)
RPS4Y1 | 201909_at
Expression level (log2)
Age
Smoker
non−smoker

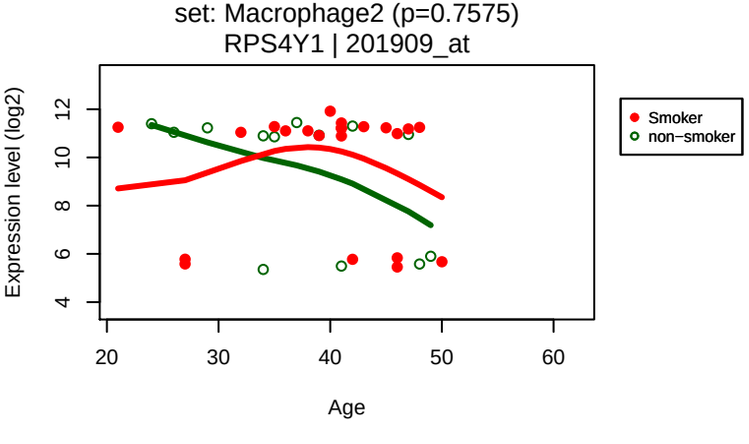

set: SmallAirway2 (p=0.5565)
RPS4Y1 | 201909_at
Expression level (log2)
Age
Smoker
non−smoker

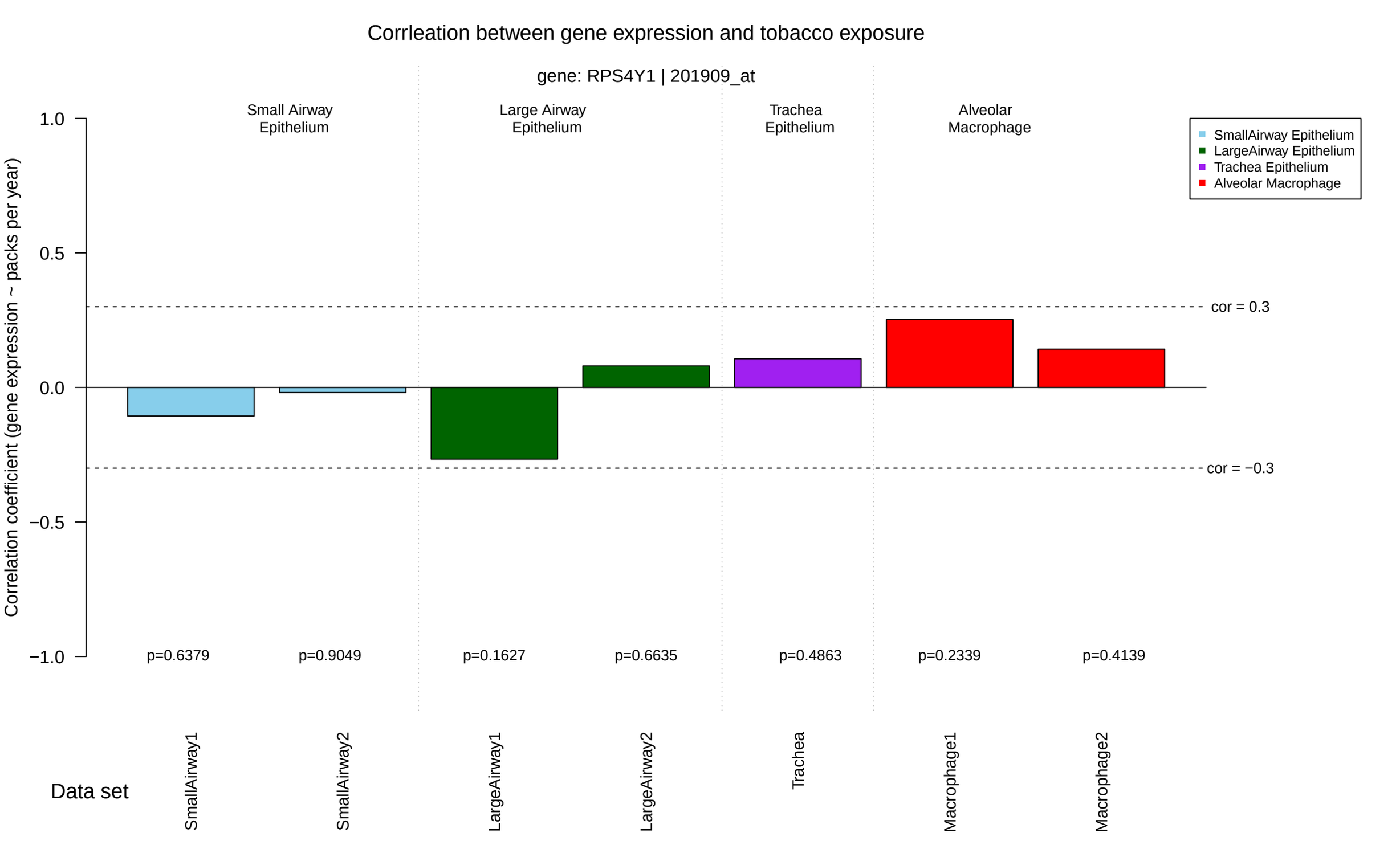

set: LargeAirway1 (p=9e−04)
RPS4Y1 | 201909_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4986)
RPS4Y1 | 201909_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3176)
RPS4Y1 | 201909_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1843)
RPS4Y1 | 201909_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7575)
RPS4Y1 | 201909_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS4Y1 | 201909_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6379
p=0.9049
p=0.1627
p=0.6635
p=0.4863
p=0.2339
p=0.4139
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2